$$\sum_{j\,:\,(i,j)\in A} f_{ij} + s_i = D_i, \qquad f_{ij},\, s_i \geq 0$$

$$\sum_{i\,:\,(i,j)\in A} f_{ij} \leq Q_j^0 + q x_j$$

$$x_j \leq M_j z_j, \quad z_j \leq x_j, \quad y_j \leq z_j$$

$$7.4 x_j \leq G_j + U_j y_j, \quad x_j + b_j^0 \leq \bar{L}_j$$

$$\sum_j (c_j x_j + F_j z_j + K_j y_j) \leq B$$

$$x_j \in \mathbb{Z}_+, \quad y_j, z_j \in \{0, 1\}$$